INCH-POUND

MIL-C-70781 (AR)
AMENDMENT 4
12 March 1993
SUPERSEDING
AMENDMENT 3
15 March 1991

MILITARY SPECIFICATION

CARTRIDGE, 60MM, SMOKE, MARKING, M722
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-70781 (AR), dated 17 October 1988, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, Change "15-12-345 - Cartridge, 60MM, Smoke, Marking, M722" to read "12902791 - Cartridge, 60MM: Smoke, WP, M722 W/Fuze, PD, M745" and make this drawing number change wherever else it appears in the specification.

Delete the following drawings:

"15-12-352 - Packing and Sealing for: 60MM Cartridge, Smoke, Marking, M722 with M745 Fuze
15-12-358 - Intermediate Packing and Marking
15-12-359 - Marking for Fiber Tube
15-12-361 - Marking for Shipping Container"

Add the following drawings:

"9293439 - Packing and Marking for Container, Ammunition, Metal PA 70, for 60MM Cartridge, HE, M720 and WP M722
9293287 - Packing and Marking Container: Ammunition, Fiber, PA 78 for 60MM Cartridge, HE, M720 and M722"

Change the drawing title: "9293440 - Box, Wirebound, Packing, Ammunition For 60MM, Cartridge , HE, M720 W/Fuze, Packing and Marking" to read "9293440 - Box, Wirebound, Packing, Ammunition For 60MM Cartridge, HE, M720 and M722, Packing and Marking."

AMSC N/A FSC 1310

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PAGE 6

4.4.3.1.b, Change "five or more duds occur" to read "four or more duds occur."

PAGE 7

Ballistic test (Proving Ground), change "126" to "135."

Change drawing number "15-12-358" to read "9293439"

PAGE 14

4.4.2.5, Change "90/56" to "135/56"

PAGE 15

4.4.2.6, Change drawing number "15-12-352" to read "9293287" and next higher assembly "15-12-358" to read "9293439"

PAGE 16

4.4.2.7, Change drawing number "15-12-352/15-12-359" to read "9293287" and next higher assembly "15-12-358" to read "9293439"

Change Minor 201 "Marking, as specified on Drawing 15-12-359 incorrect or illegible" to read "Marking missing, misleading or unidentifiable."

PAGE 17

* 4.4.2.8, Major 101, 102, 103 and 104: Change requirement paragraph from "5.2" to "5.1"

Change drawing number "15-12-358" to read "9293439"

* Add new Minor 201 as follows:

"201 Graphical firing scale not present in every metal container ...0.65% ...5.1 ...Visual."

PAGE 18

4.4.2.9, Change drawing number "15-12-358" to read "9293439"

* Major 101 and Minor 202, change requirement paragraph from "5.2" to"5.1"

* Minor 201, Change requirement paragraph from "5.4.2" to "5.3"

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PAGE 19

4.4.2.10, Change drawing number "9293440/15-12-361" to read "9293440"

* Minor 201 and 202, Change requirement paragraph from "5.3" to "5.2"

* Minor 203, Change requirement paragraph from "5.4" to "5.3"

PAGE 20

4.4.3.1.1, Change "90" to "135"

4.4.3.1.1.b, Change "Two or more duds occur" to read "Four or more duds occur."

4.4.3.1.2, Change "90" to read "56."

4.4.3.1.2.b, Change "two or more duds occur" to read "three or more duds occur."

PAGE 21-22

4.5.3.2, Delete in its entirety and substitute the following:

"4.5.3.2 Production test. The cartridges for production lot acceptance shall be conditioned at 70 degrees F ± 2.5 degrees F. The sample for the first three production lots (135) shall be separated into three equal groups; one group tested at charge zero, one group tested at charge two, and one group tested at charge four. After three consecutive lots have complied with the acceptance requirements, the production lot sample (56) shall be separated into two groups; one group tested at charge zero, and one group tested at charge four. All cartridges from the ballistic sample shall be radiographed prior to shipment to the ballistic test facility. Film from acceptable lots may be discarded. The cartridge shall be fired from a standard 60MM Lightweight Company Mortar System (LWCMS) mortar at an elevation of 45 degrees from a hard mount (a solid base test fixture) within two minutes after removal from the conditioning chamber. The individual velocities, pressures, deflections, and ranges shall be recorded on the firing record. Observation shall be made for misfires and duds, and shall be recorded. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and

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arithmetic averages shall be calculated and recorded. Range probable error shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded. Hawk radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of these rounds, and prints of the preceding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded. A smear camera shall be used to photograph all test samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all the other rounds in the particular charge zone, shall be attached to the firing record. Film from acceptable lots may be discarded."

PAGE 24

5.1, Change to read, "Packaging. Packaging shall be as specified on drawings (dwgs.) 9293287 and 9293439."

5.1.1, Delete in its entirety.

5.1.2, Delete in its entirety.

5.1.3, Delete in its entirety.

PAGE 25

5.2, Change to read "Packing. Packing shall be as specified on dwg. 9293440."

5.2.1, Delete in its entirety.

5.2.2, Delete in its entirety.

5.2.3, Delete in its entirety.

5.3, Change to read "Marking. Marking shall be as specified on dwgs. 9293287, 9293439 and 9293440."

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5.3.1, Delete in its entirety.

5.3.2, Delete in its entirety.

5.3.3, Delete in its entirety.

6.3, Add: "Commander, U.S. Army, AMCCOM, ATTN: AMSMC-QAC-M (A), Aberdeen, MD 21010-5423."

NOTE: The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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